# Recreation of a U-Net: Convolutional Networks for Biomedical Image Segmentation (2015) by *Olaf Ronneberger, Philipp Fischer and Thomas Brox*

Ksawery Kochanowicz, Miloš Karapandžić, Denis Dubovicky

GROUP 7

# Contents

**1** Introduction & motivation ..... 2
**2** The problem addressed in the paper ..... 2
3 Background ..... 2
4 Methodology ..... 3
4.1 Dataset and data preprocessing ..... 3
4.2 Model Architecture ..... 4
4.2.1 Encoder ..... 4
4.2.2 Bottleneck ..... 4
4.2.3 Decoder ..... 4
4.2.4 Output Layer ..... 4
4.3 Training Procedure ..... 4
4.4 Results & analysis ..... 4
4.5 Discussion ..... 5

# 1 Introduction & motivation

Our choice for the paper we want to reproduce came out to preference. All members of our team were already interested in image segmentation beforehand. We also realized that Image classification with neural networks (CNNs for image analysis), were a huge part of the curriculum for the Applied Computer Vision subject, all three of us are going to be attending in the 4th semester. All of those reasons have caused us to research this topic, in order to have a better grasp of its basics. That's when we encounter the research paper about Biomedical Image Segmentation using a U-Net.

# 2 The problem addressed in the paper

Recently, deep convolutional networks have become more and more efficient with visual recognition tasks. A typical convolutional network results in an output of a single class label. In some fields, such as biomedical image processing require a different approach: class label should be assigned to each pixel of the image. Another downside of a typical convolutional network is the fact that it required a large amount of training data, which is unachievable in case of many biomedical tasks. The goal of the paper was to resolve those issues.

The paper claims to presents a "more elegant architecture" than the standard convolutional network. The authors aimed to improve the precision of the segmentation, by adding successive layers to the contracting networks, and then replacing pooling operators with upsampling operators, which makes the layers increase the resolution of the output.

# 3 Background

Earlier CNN models, for example, the one introduced by Krizhevsky et al., have showed a substantial breakthrough in image classification, achieved by training deep networks on large datasets like ImageNet. Those architectures show great efficiency at assigning single label to an entire image, they don't provide pixel-level localization, which as mentioned before, is a necessity in fields like biomedical imaging.

To tackle this issue, Ciresan et al. proposed a different approach – a sliding-window CNN, that predicts labels for individual pixels based on local patches. This method proved to be effective but also flawed. Since the network must process many overlapping patches, it results in a High computational cost. Another issue is that the small patches lack global context, and large patches require polling layers that reduce spatial precision.

In more recent research, multi-layer feature combinations were explored, to improve localization, while retaining context, but these approaches were still lacking in case of architectural elegance and efficiency.

Then came the fully convolutional networks (FCNs), which introduced a new model by completely replacing fully connected layers with upsampling layers, which enabled end-to-end pixel-wise prediction. This architecture, extended in the original paper build on the FCNs by adding:

- Skip to connections in order to combine high-resolution and contextual features,
- Symmetric down sampling and upsampling paths,
- Large feature channel capacity in the expansive path to preserve context,
- Seamless tiling for segmenting arbitrarily large images,
- Data augmentation via elastic deformations to overcome limited training data.

Additionally, the authors introduced a weighted loss function, to better separate the touching objects-and important challenge in cell and tissue segmentation. This architectural foundation resulted in state-of-the-art performance in several biomedical segmentation competitions, demonstrating both accuracy and practicality in low-data scenarios.

# 4 Methodology

The objective of this study was to reproduce the core architectural principles of the original 2015 U-Net and evaluate its effectiveness on a biomedical image segmentation task. Performance was assessed using the Intersection over Union (IoU) metric and compared against a baseline patch-based CNN model to verify the benefits of the encoder–decoder structure and skip connections characteristic of U-Net.

## 4.1 Dataset and data preprocessing

The dataset consisted of 1,312 ultrasound images including their binary tumour masks. The images, published in 2018 by Arya Shah under a public domain license, depict benign and malignant breast tumours and are provided in 500×500-pixel PNG format. All files were imported into Google Colab via the Kaggle API and programmatically scanned to match each image with its corresponding ground-truth mask, identifiable by the “mask” keyword in the filename.

To meet computational constraints and align with common U-Net reproduction practices, all images were converted from RGB to grayscale and resized to 128×128 pixels. Each image was normalized to the [0,1] range to stabilize gradient updates during training. The final input shape was defined as (128, 128, 1). Masks were ensured to be binary and matched with their corresponding images. A validation split of 10 percent was used during training.

## 4.2 Model Architecture

### 4.2.1 Encoder

The encoder comprised four downsampling blocks. Each block followed the canonical U-Net configuration: two convolutional layers with 3×3 kernels, ReLU activation, padding to preserve spatial resolution, and He normal weight initialization. A 2×2 max-pooling layer reduced spatial dimensions by a factor of two at each stage. Feature channels doubled progressively from 64 to 512, enabling extraction of increasingly abstract representations.

### 4.2.2 Bottleneck

The bottleneck formed the deepest part of the network and consisted of two 3×3 convolutional layers with ReLu activation function and He distribution initialization. No pooling was applied. This stage captured the highest-level semantic structure before reconstruction in the decoder.

### 4.2.3 Decoder

The decoder mirrored the encoder with four upsampling blocks. Each block began with a 2×2 transposed convolution with stride of 2 to restore spatial resolution. The upsampled output was concatenated with the corresponding encoder feature map via skip connections, recovering spatial detail lost during pooling. Each concatenation was followed by two 3×3 convolutional layers with ReLU activation, padding, and He initialization. The number of feature channels decreased symmetrically toward the output.

### 4.2.4 Output Layer

A final 1×1 convolution with a sigmoid activation produced a single-channel probability map, estimating the likelihood of each pixel belonging to tumour tissue or background.

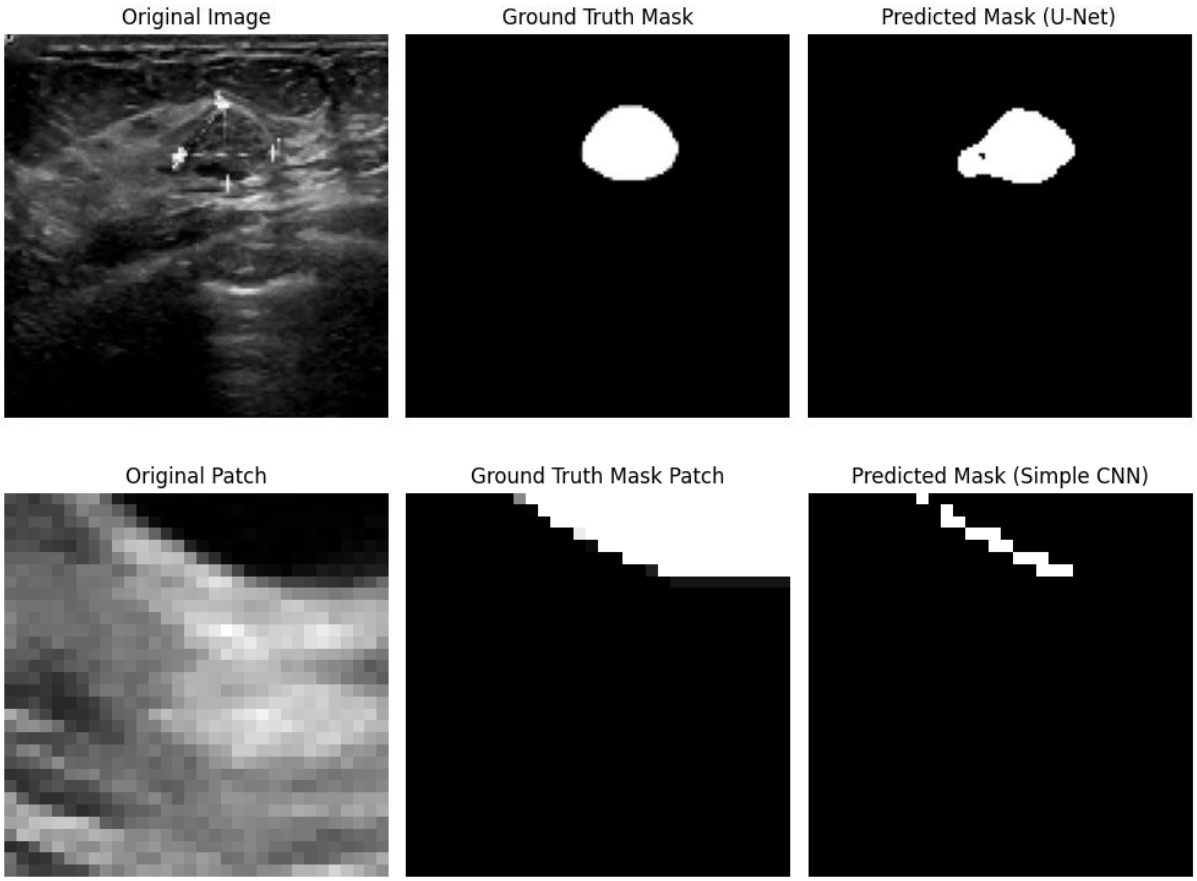
## 4.3 Training Procedure

The model was compiled for binary semantic segmentation using the Adam optimizer and binary cross-entropy loss function. IoU was monitored as the primary performance metric. Training was conducted for 50 epochs with a batch size determined dynamically based on memory limits, using a 90/10 train-validation split. Verbose logging was enabled to track loss and metric progression throughout training.

## 4.4 Results & analysis

The U-Net model achieved an IoU score of **0.5935**, while the simpler CNN model achieved a significantly lower IoU of **0.0032**. This stark difference indicates that the U-Net architecture is far more effective for image segmentation in this context. Visual analysis confirms that U-Net model's predictions fairly match the ground truth masks, accurately segmenting the breast tumours. The simple batch sampling CNN model's predicted mask (see plots) failed to identify numerous pixels as part of the mask. This

suggests that the simple CNN model either learned very little or was unable to capture the complex features required for this segmentation task. It can be due to its simpler architecture and lack of skip connections.




In conclusion, for this specific image segmentation task, the U-Net model demonstrates superior performance in both quantitative (IoU score) and qualitative (visual prediction) aspects compared to the simpler benchmark model. Thus, confirming the advantages and effectiveness proposed in the original study.

## 4.5 Discussion

In this project we were able to recreate a U-Net model and see why it works well for biomedical image segmentation. The skip connections clearly helped the model keep important details, and even with a fairly small dataset the U-Net still learned decent segmentations after some training. As of the biggest limitations in this project, we want to mention the Google Collab computational power. We had no prior knowledge about computer vision, due to that fact we also didn't know how well this project will be running in it. Additionally, computer vision as a concept was totally new and this project introduced us to it. Overall, we learned that the U-Net architecture is actually quite intuitive once you understand how the encoder and decoder work together, and it definitely performed better than our patch-based baseline model.